EXPERIMENT – 4

String and Block Data Operations Using 8086 Microprocessor

Objective

To develop and execute 8086 assembly language programs for **block data transfer, string comparison, character searching and case conversion** using string instructions such as MOVSB, CMPSB, SCASB, LODSB and STOSB with repeat prefixes.

Algorithm

1. Initialize the source and destination addresses using SI and DI.
2. Load the number of bytes into CX.
3. Use REP MOVSB to transfer the block of data.
4. Use REPE CMPSB to compare two strings.
5. Use REPNE SCASB to search for a character.
6. Use LODSB and STOSB to process and convert string characters.
7. Store the results in the specified memory locations.
8. Verify the results with the expected values.

Software and Hardware Required

- 16-bit Emulator (EMU8086 / MASM)
- 8-bit / 16-bit Processor Trainer Kit
- Keyboard
- Power Supply

A) Block Transfer Using REP MOVSB

Code

```
CLD
MOV SI,3000H
MOV DI,3100H
MOV CX,000BH
REP MOVSB
STD
MOV SI,300AH
```

MOV DI,300DH

MOV CX,000BH

REP MOVSB

CLD

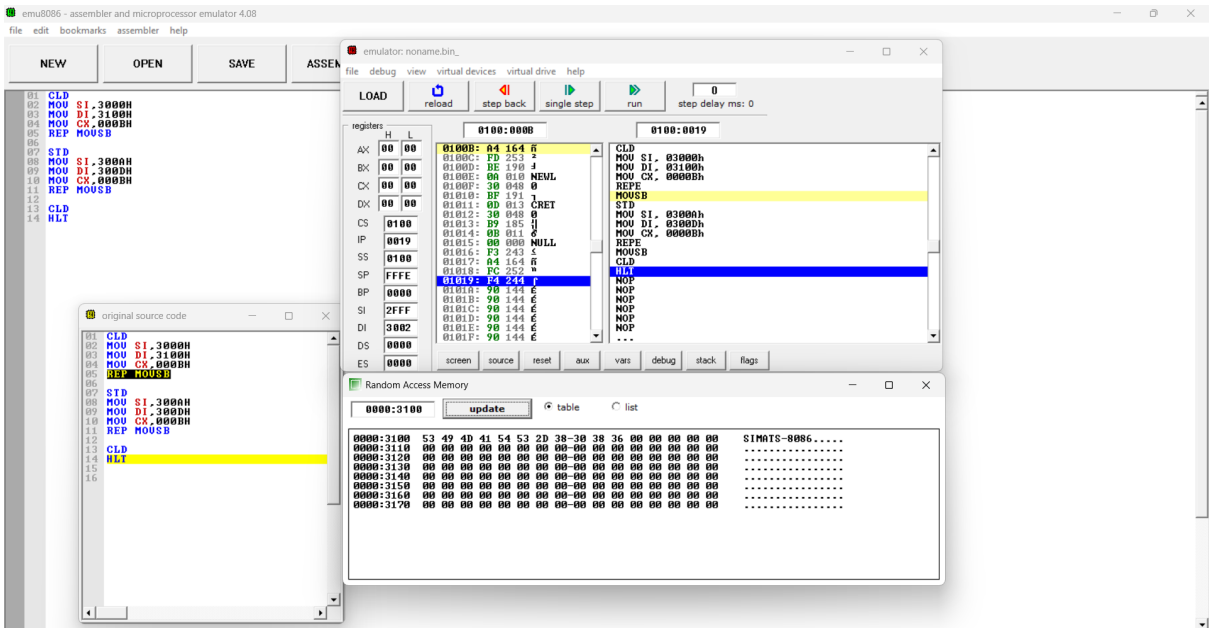
HLT

Input

String: **SIMATS-8086**

Address	**Data**
3000H–300AH	53 49 4D 41 54 53 2D 38 30 38 36

Output



Address	**Result**
3100H–310AH	SIMATS-8086
3000H–300DH	SIMSIMATS-8086

The master record specifies the first forward copy and then the overlapping move using DF =1.

B) String Comparison Using REPE CMPSB

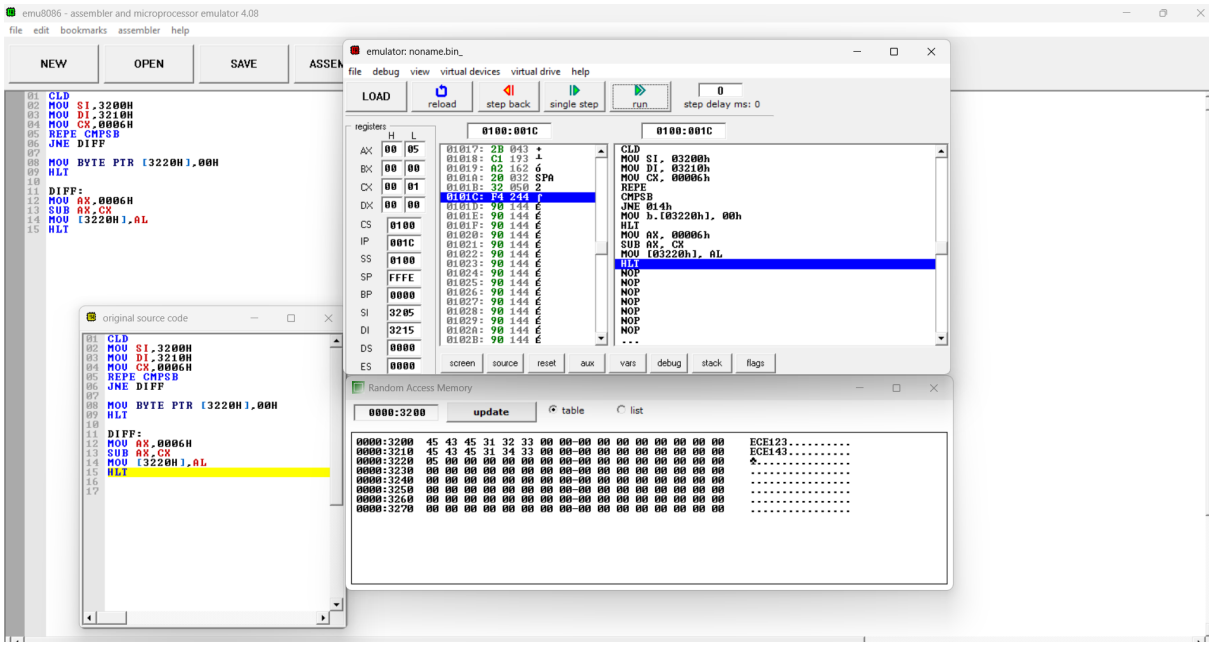
Code

```
CLD
MOV SI,3200H
MOV DI,3210H
MOV CX,0006H
REPE CMPSB
MOV AX,0006H
SUB AX,CX
MOV [3220H],AL
HLT
```

Input

Address	String
3200H	ECE123
3210H	ECE143

Output



Address	Data
3220H	05H

Result: First mismatch occurs at the **5th character**.

C) Character Search Using REPNE SCASB

Code

CLD

MOV DI,3300H

MOV CX,000CH

MOV AL,41H

REPNE SCASB

MOV AX,000CH

SUB AX,CX

DEC AX

MOV [3311H],AL

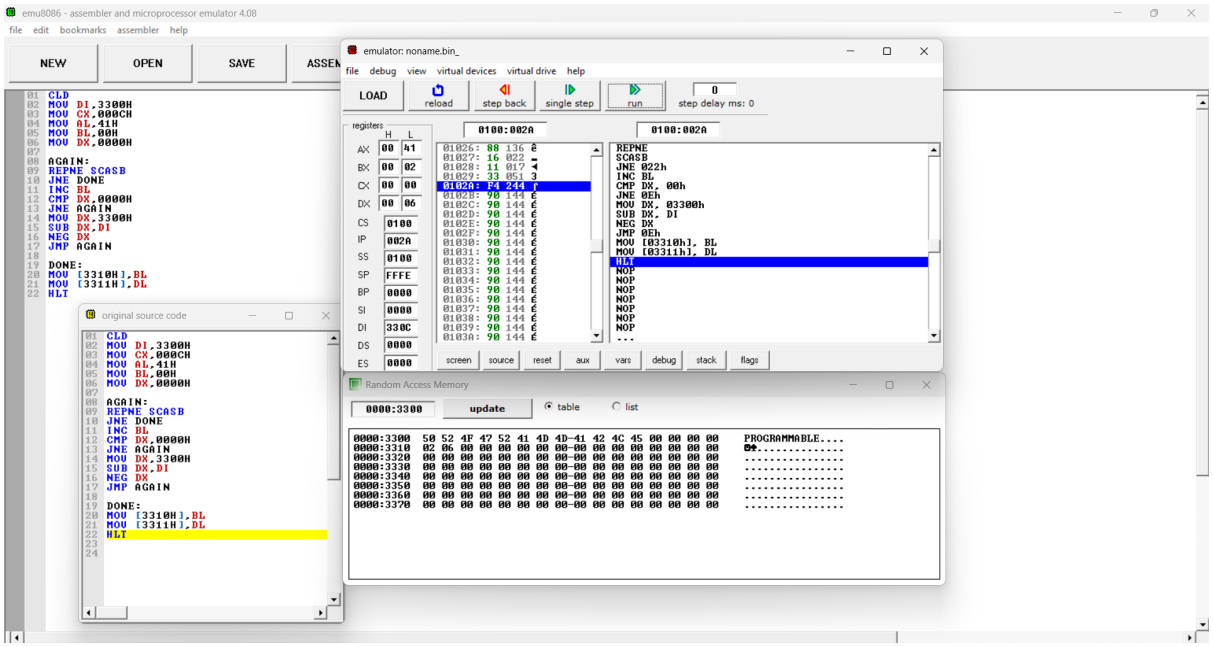
HLT

Input

String: **PROGRAMMABLE**

Search character: **A**

Output



Address	Data
3310H	02H
3311H	06H

Result:

- Number of occurrences of **A = 02H**
- First occurrence position = **06H**

D) CASE CONVERSION USING LODSB AND STOSB

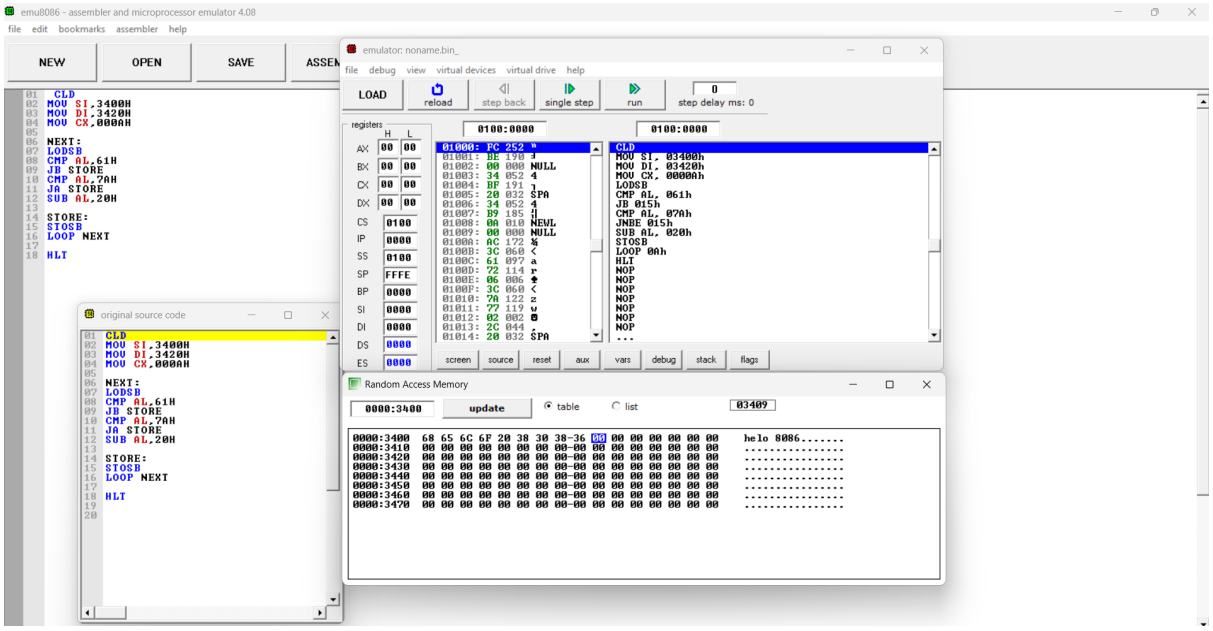
CODE

```
CLD
MOV SI,3400H
MOV DI,3420H
MOV CX,000AH
NEXT:
lodsb
CMP AL,'a'
JB STORE
CMP AL,'z'
JA STORE
SUB AL,20H
STORE:
STOSB
LOOP NEXT
HLT
```

Input

hello 8086

Output



HELLO 8086

Address	Data
3420H onwards	HELLO 8086

The master record specifies "hello 8086" as the input and "HELLO 8086" as the expected converted string.

Result

Thus, the **string comparison, block transfer, character searching and case conversion operations were successfully performed using 8086 string instructions**. The obtained results were verified with the expected memory contents.

Student Reflection

I learned how to perform string and block data operations using the 8086 microprocessor. I understood the use of SI, DI and CX registers along with MOVSB, CMPSB, SCASB, LODSB and STOSB. I also learned how the direction flag and repeat prefixes are used for efficient string processing.